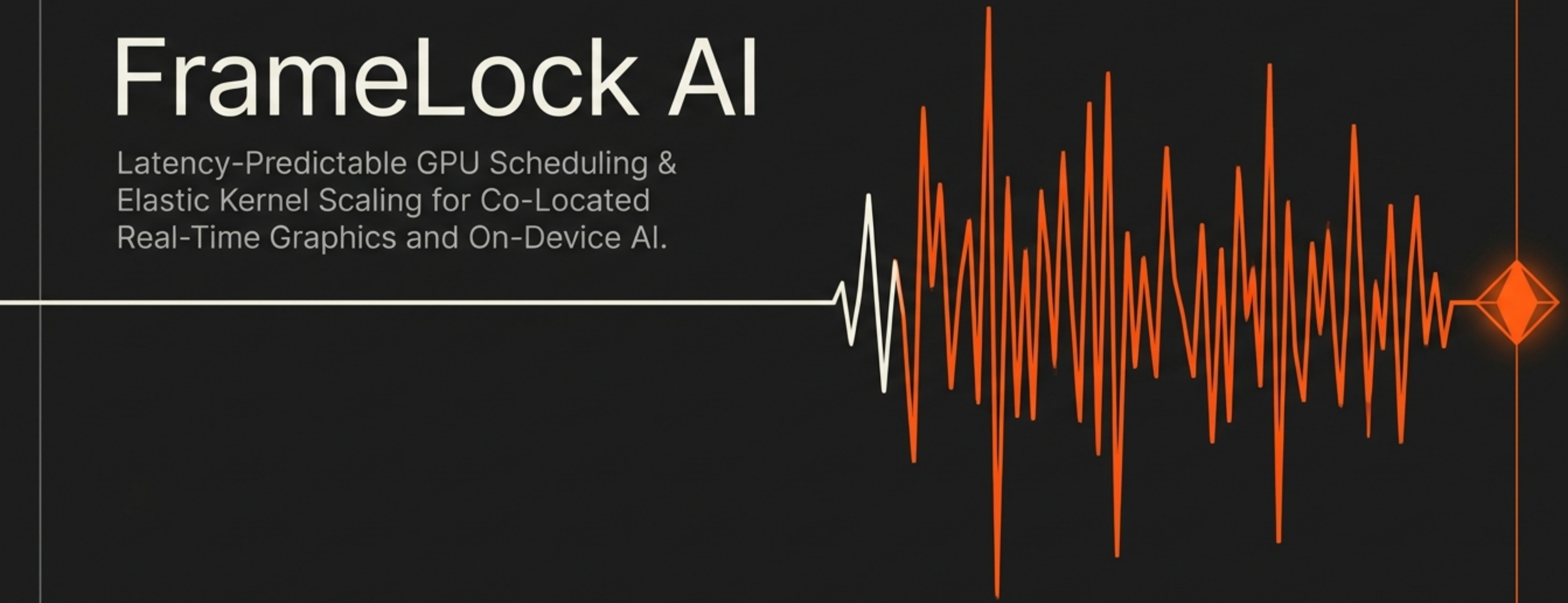
FrameLock AI

Latency-Predictable GPU Scheduling & Elastic Kernel Scaling for Co-Located Real-Time Graphics and On-Device AI.



>_ SYS_ARCHITECT: Sreenandh S.

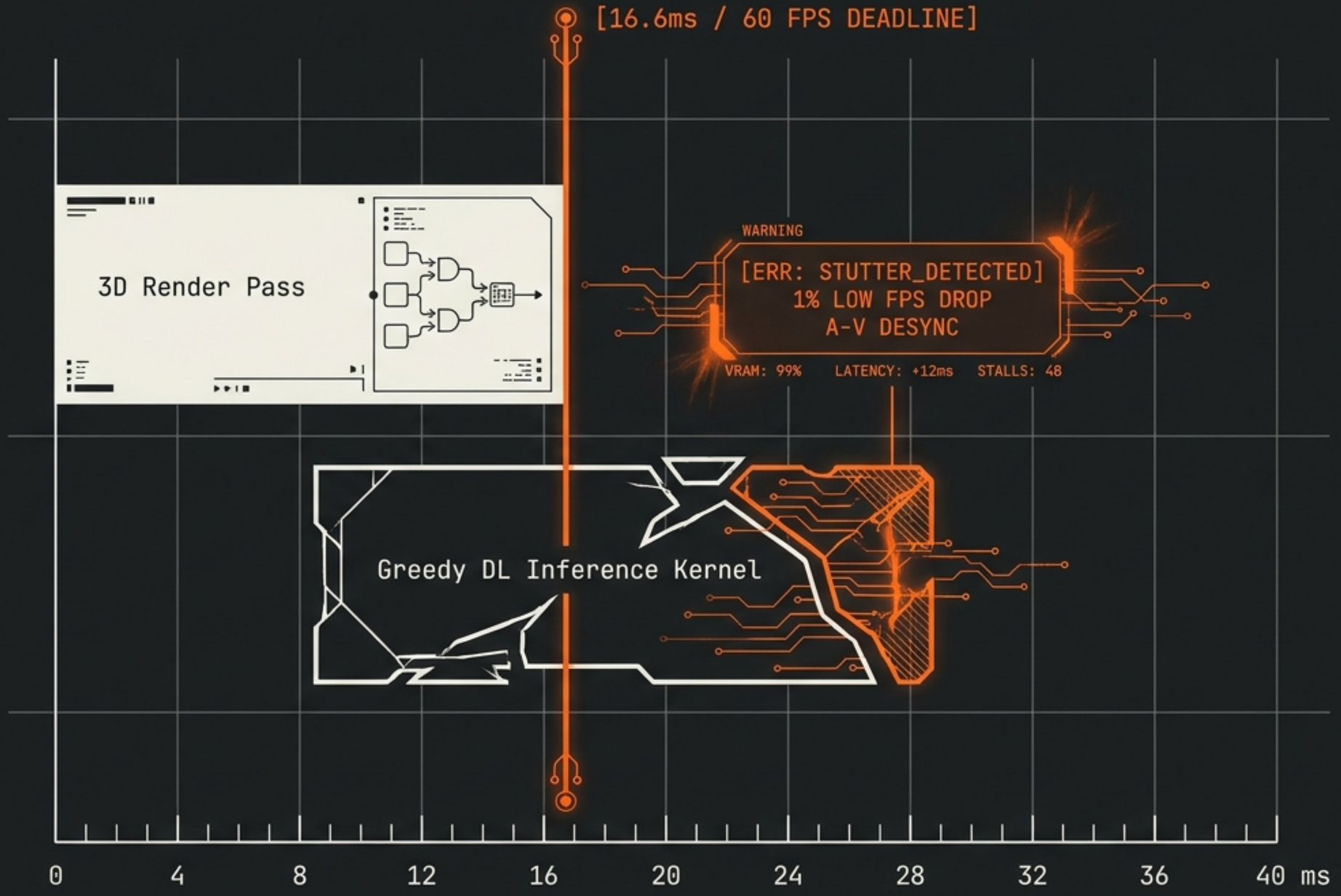
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The Contention Bottleneck

Modern 3D graphics (game engines, VR, robotics) enforce a rigid 16.6ms deadline.

Standard AI inference kernels are burst-heavy and ignorant of this constraint.

Result:
Severe VRAM contention and pipeline stalling. Unsolvable by standard OS scheduling.



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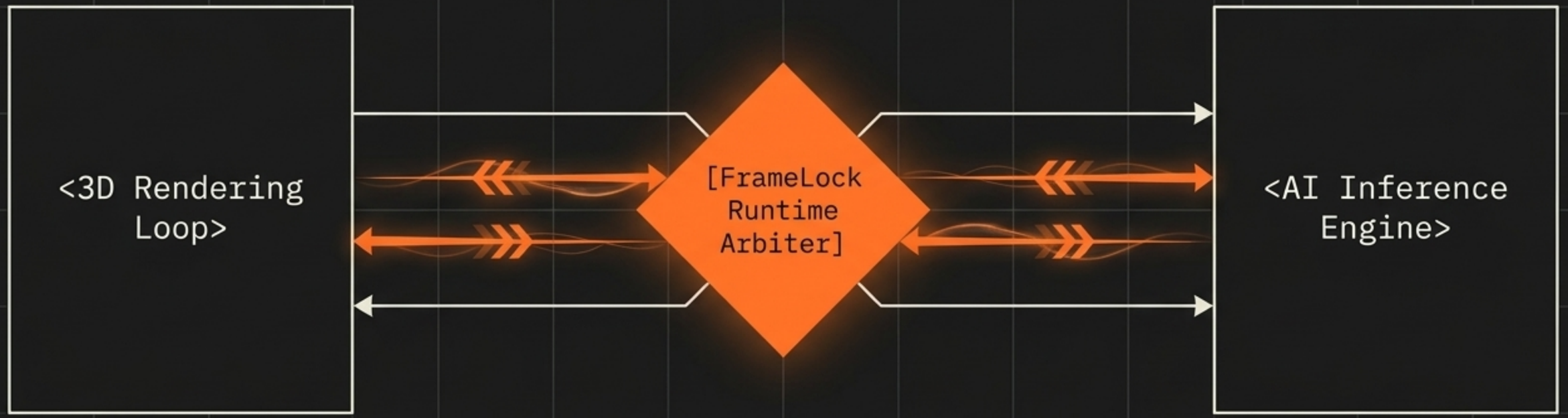
The Diagnostic Contention Matrix

Solution	Resolves VRAM Starvation?	Handles Pipeline Thrashing?	Continuous Execution?
NVIDIA NVIGI (SDK) Rigid manual developer toggles.	○	○	○
VUDA (SJTU) Partitions cores, ignores bandwidth.	✕	○	○
GPUSched (IEEE) Slices AI for non-existent idle gaps.	○	○	✕
FrameLock AI	✓	✓	✓

System Takeaway: Current fixes merely pause or sacrifice workloads. They do not solve shared resource contention at the memory bus.

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FrameLock: The Arbiter Architecture



- **Intelligent Arbiter**

  An intelligent, dynamic GPU runtime layer operating at the millisecond scale.

- **Real-Time Headroom**

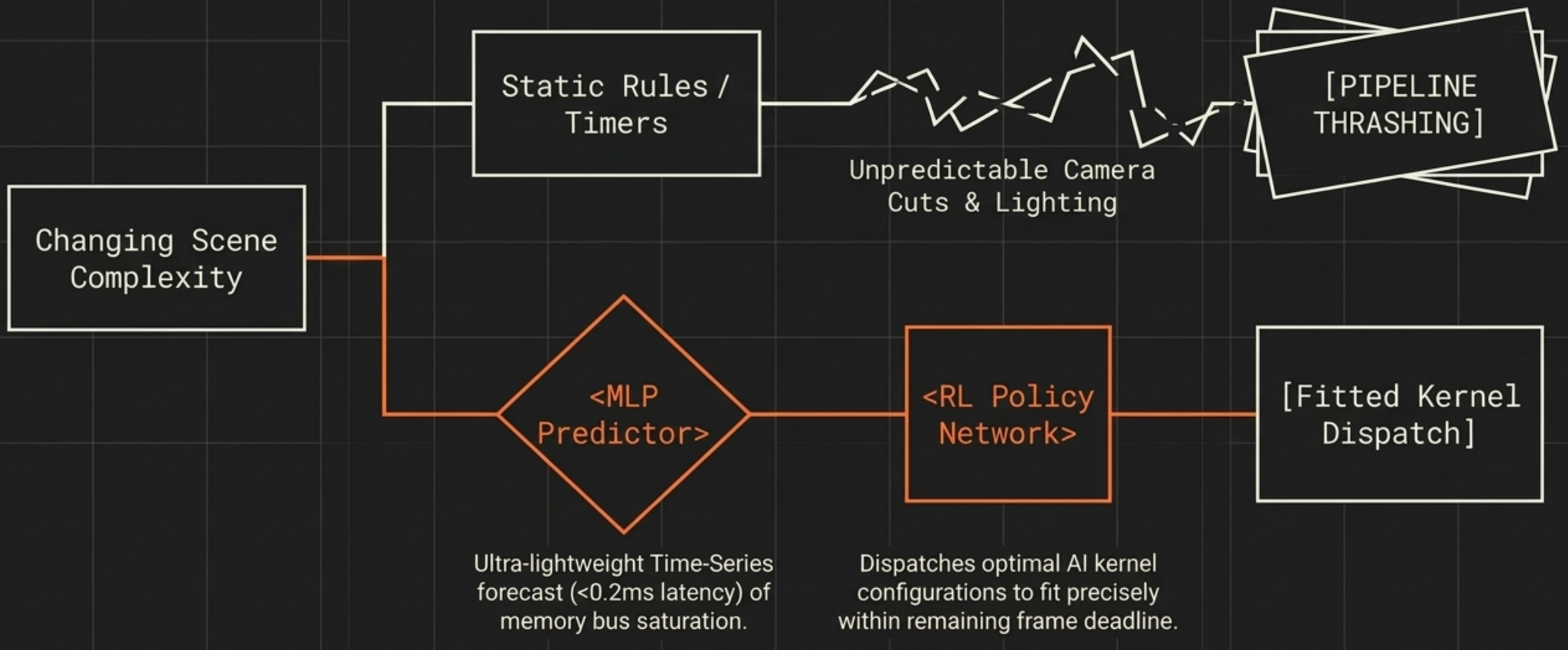
  Actively reads frame-pacing headroom in real time and automatically scales the operational intensity of the AI model.

- **Continuous Intelligence**

  Local intelligence runs continuously without stealing required memory bandwidth from the primary graphics renderer.

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Algorithmic Architecture: Why Static Rules Fail



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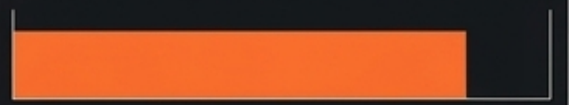
[PROBE 01] L2 Cache Hit/Miss Rates

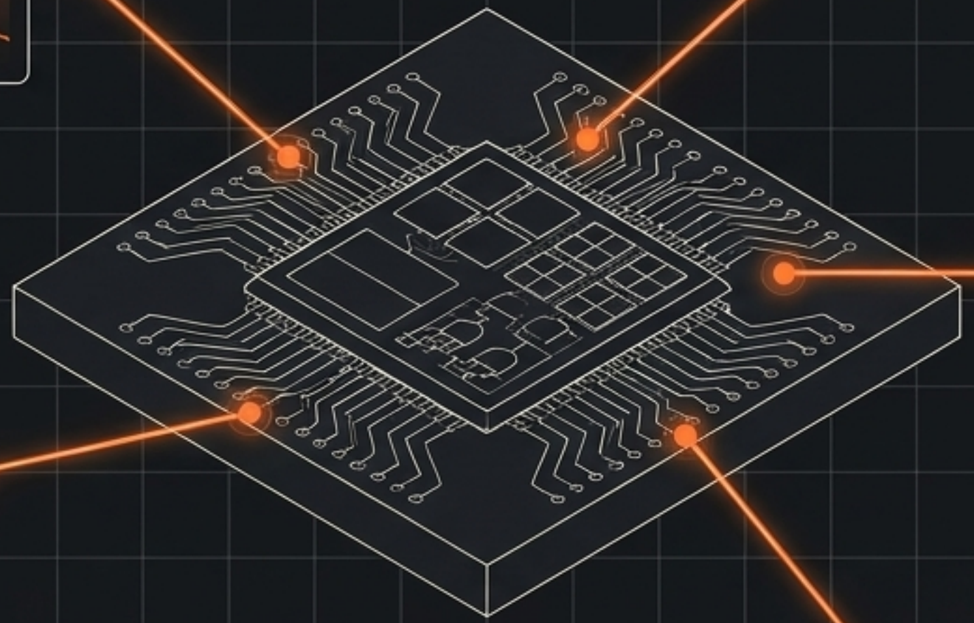
Hit Rate: 92.45% | Miss Rate: 7.55%



[PROBE 02] GPU Memory Bus Bandwidth Utilization

Peak BW: 890.2 GB/s | Util: 85.3%





[PROBE 03] DirectX/Vulkan Draw Call Counts

Frame Calls: 12,450 | Peak Rate: 15.2k /s

12,450

[PROBE 05] CUDA/Triton Kernel Execution Durations

Avg Exec: 0.45ms | Max Exec: 1.2ms



[PROBE 04] Camera Angular Velocity

Yaw: 1.25 deg/s | Pitch: 0.10 deg/s

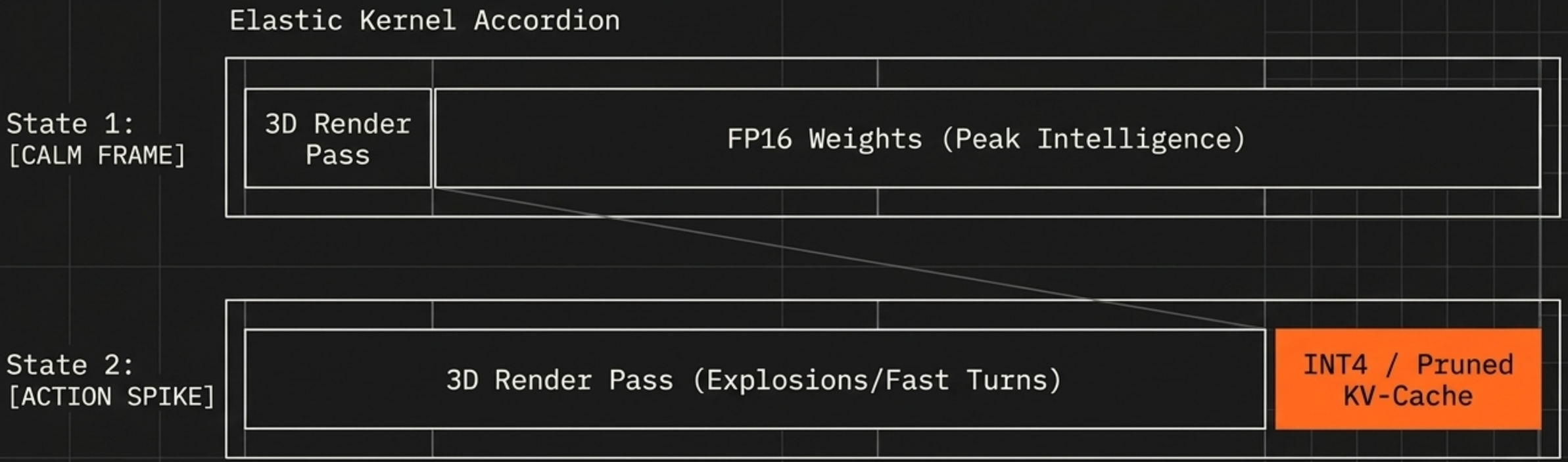


Data Independence & Privacy:

System input relies exclusively on low-level hardware performance counters and rendering metrics. Zero user/PII data is required. Trained exclusively on synthetic graphics test scenes and standard benchmarks.

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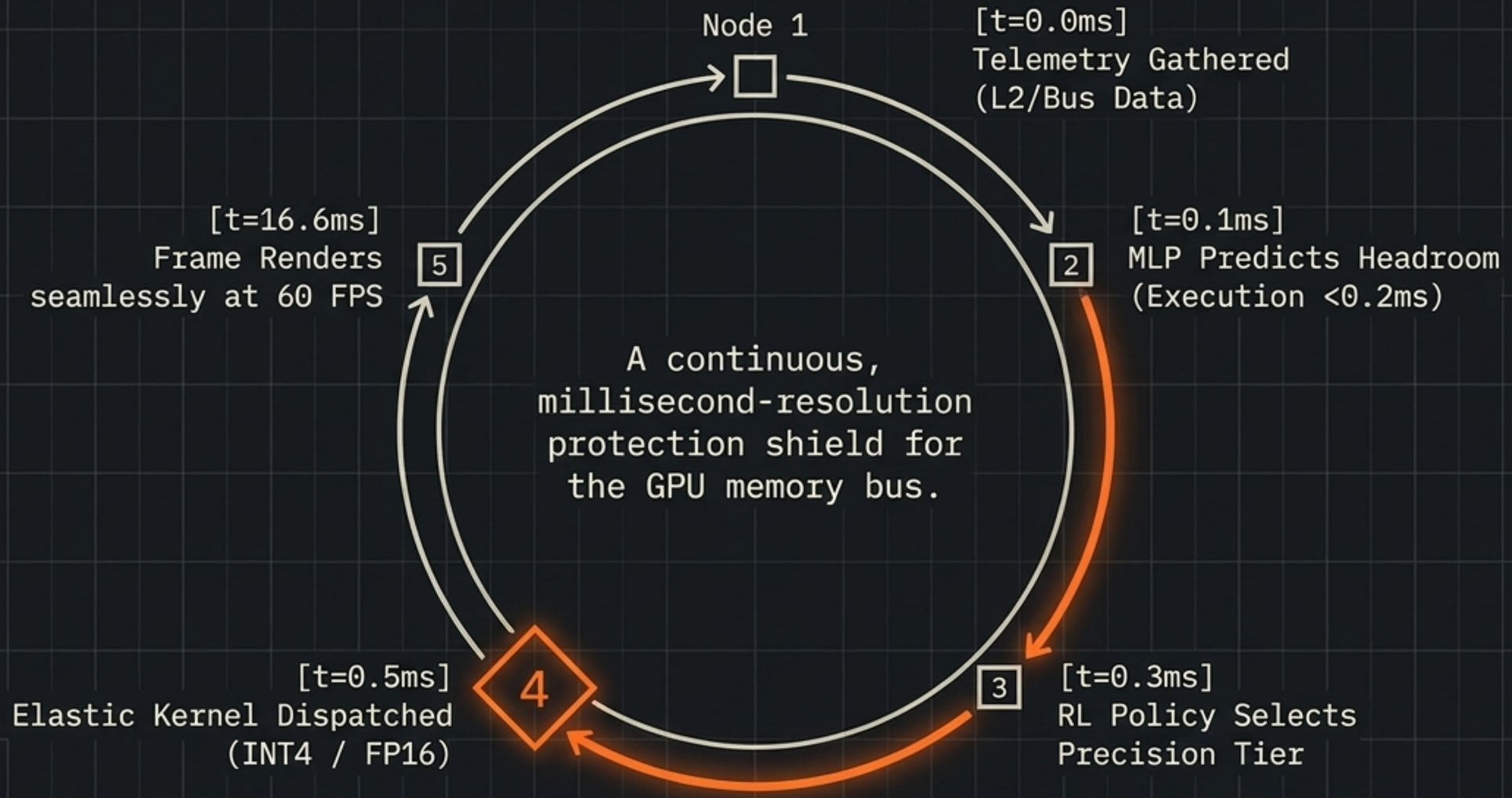
The Breakthrough: Bandwidth-Aware Elastic Kernel Switching



Result: Seamlessly switches mid-inference to ultra-lean INT4 kernels or early-exit layers. Guarantees 60 FPS while keeping local AI streaming responses without interruption.

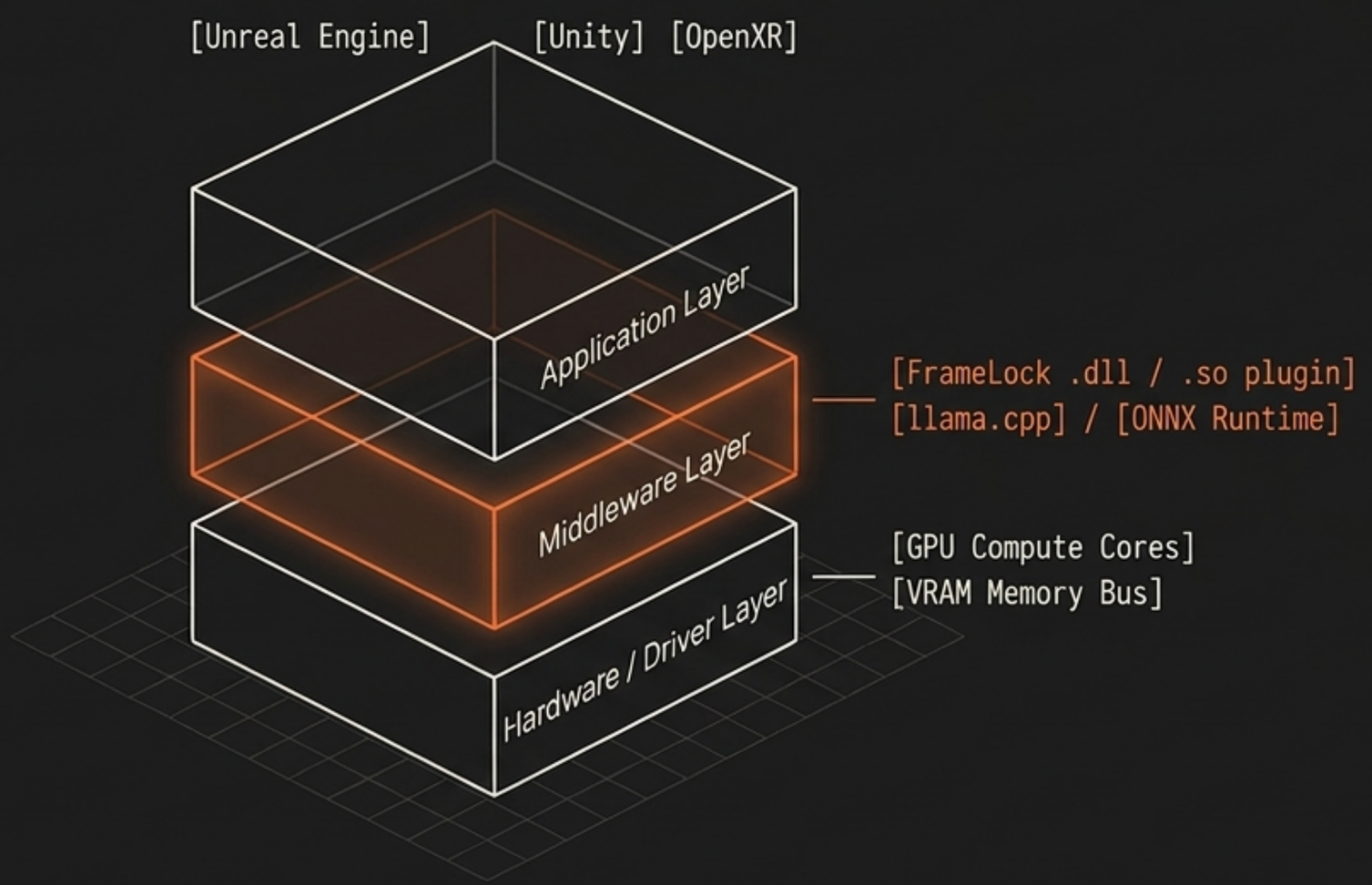
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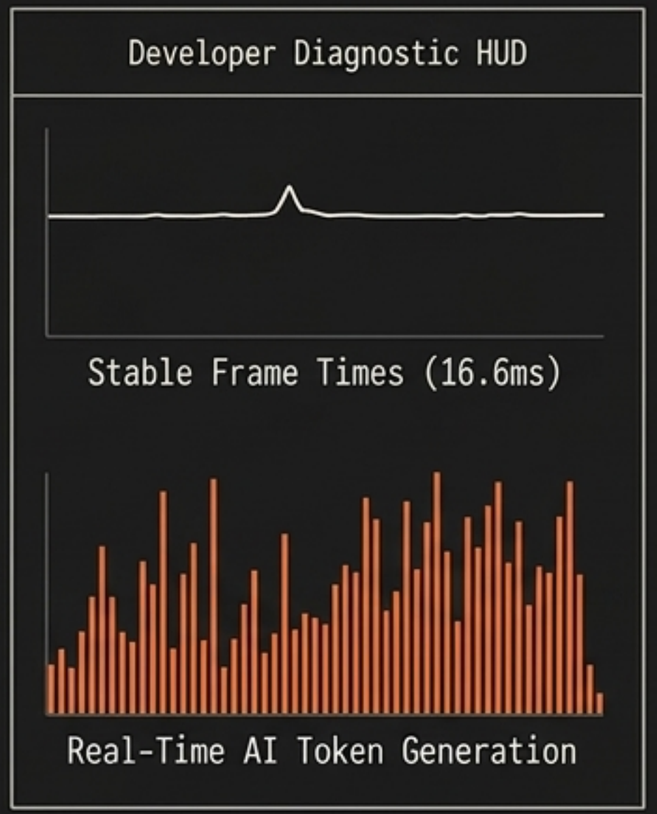
The Closed-Loop Arbiter



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Architecture & Deployment Stack





Deployed as a lightweight C++/CUDA runtime library compatible with major graphics platforms.

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Key Challenges & Active Research Vectors

[SYS_WARN_01]: Predictor Overhead

ML telemetry prediction must execute fast enough that it does not consume the exact frame budget it seeks to protect.

[SYS_WARN_02]: Driver Black Boxes

Navigating undocumented hardware scheduling behavior deeply embedded in proprietary NVIDIA and AMD display drivers.

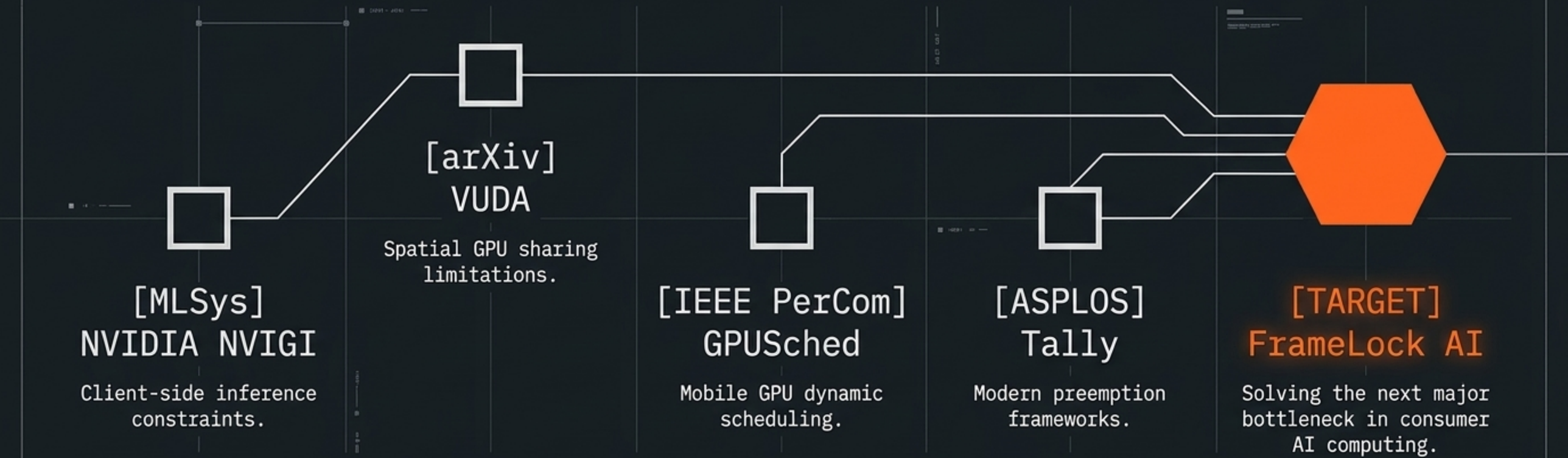
[SYS_WARN_03]: Kernel Transition Latency

Minimizing GPU pipeline bubbles when hot-swapping between different quantization levels (FP16 ↔ INT4) mid-inference.

Transparent analysis of system friction points ensures architectural integrity.

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Anchored in Peer-Reviewed Systems Architecture Literature



Built directly upon verified system limitations to define the future of co-located GPU computing.

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